

COMP 302 Software Engineering Game Project Requirements KU Royale

Fall 2025 - Phase 2

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A Multi-Player Strategy Battle Game

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1. Phase 2: Advanced Features and Multiplayer

Phase 2 Overview:

Phase 2 builds upon the foundation of Phase 1 by introducing multiplayer capabilities, progression systems, and advanced gameplay features. This phase emphasizes software design patterns and architectural principles while adding significant new functionality.

1.1. Design Patterns Requirement

In Phase 2, you must demonstrate mastery of software design patterns. Each pattern should be:

- Properly documented with comments explaining its purpose
- Justified in your design pattern report (why this pattern was chosen)
- Illustrated with UML diagrams in your submissions

Note: You do not need to know which specific patterns to use beforehand. The features described below naturally lend themselves to certain patterns, and you should discover and apply appropriate patterns as you design and implement each feature.

1.2. Phase 2 Features Overview

Phase 2 introduces six major features:

1. **Local Player vs Player Mode** - Two human players on the same device
2. **Network Multiplayer** - Play over LAN or internet
3. **Card Evolution & Rarity System** - Upgrade cards and tiered rarity
4. **Challenge Mode** - Pre-designed scenarios with specific objectives
5. **Quest and Achievement System** - Daily quests and permanent achievements
6. **Card Combo System** - Synergy bonuses for card combinations

2. Feature 1: Local Player vs Player Mode

2.1. Functional Requirements

Implement a local multiplayer mode where two human players can battle on the same device.

Core Requirements:

- Two human players compete against each other
- Each player has their own deck (8 cards each)
- Each player has independent Elixir counter (starts at 5, max 10)
- Players control their respective sides of the arena
- Support either turn-based or simultaneous card placement (your choice)
- Same win conditions as Phase 1 (King Tower destruction or Crown Tower count)
- Both players can see the entire battlefield

UI Requirements:

- Main menu option: "Local PvP"
- Deck selection screen for both players
- Clear visual indication of which player's turn it is (if turn-based)
- Separate Elixir displays for both players
- Victory screen showing which player won

3. Feature 2: Network Multiplayer

3.1. Functional Requirements

Enable players to battle over a network connection (LAN or internet).

Connection Architecture:

- **Host-Client model:** One player hosts (acts as server), other player joins (acts as client)
- Host specifies a port number (default: 8080)
- Client connects using host's IP address and port
- Connection status indicator showing "Connected" / "Connecting" / "Disconnected"

Synchronization Requirements:

The following game events must be synchronized between players:

- Card placements (which card, where placed, timestamp)
- Unit movements and attacks
- Tower health updates

- Elixir regeneration
- Match timer
- Victory/defeat conditions

Network Lobby:

Before the match starts, display a lobby screen showing:

- Both players' names
- Deck preview (card names only, not full stats)
- Ready/Not Ready status for each player
- "Start Match" button (enabled only when both players are ready)
- Connection quality indicator

Disconnection Handling:

- If a player disconnects during a match:
 - Show "Opponent Disconnected" message
 - Attempt reconnection for 5 seconds
 - If reconnection fails, award victory to the remaining player
- Graceful shutdown: notify opponent before closing connection

Network Protocol:

You must design a simple message protocol. Suggested format:

MESSAGE_TYPE|player_id|data|timestamp

Examples:

CARD_PLACED|1|Knight|5.2,3.8|00:45

TOWER_DAMAGED|2|CrownLeft|450|01:23

ELIXIR_UPDATE|1|7|01:24

Technical Notes:

- Use Java's `Socket` and `ServerSocket` classes
- Consider threading for non-blocking network I/O
- Test on localhost (127.0.0.1) before testing over network

4. Feature 3: Card Evolution & Rarity System

4.1. Functional Requirements

Implement a progression system where players can upgrade cards and cards have different rarity levels.

4.1.1. Card Levels

All 28 cards can be upgraded through 3 levels:

- **Level 1:** Base stats (Phase 1 values)
- **Level 2:** +10% HP and +10% DMG
- **Level 3:** +20% HP and +20% DMG

Example: A Level 1 Knight has 600 HP and 75 DMG. At Level 2: 660 HP and 82.5 DMG (round to 83). At Level 3: 720 HP and 90 DMG.

4.1.2. Card Rarity System

Classify all 28 cards into 4 rarity tiers:

Rarity	Cards (12 total)
Common	Skeletons, Goblins, Spear Goblins, Archers, Knight, Bomber, Cannon, Tombstone, Arrows, Zap

Rarity	Cards (10 total)
Rare	Musketeer, Mini P.E.K.K.A, Valkyrie, Minions, Barbarians, Tesla, Mortar, Goblin Hut, Fireball

Rarity	Cards (4 total)
Epic	Giant, Wizard, Bomb Tower, Inferno Tower

Rarity	Cards (2 total)
Legendary	Hog Rider, Minion Horde

4.1.3. Economy System

Players earn gold currency after each match:

- **Victory:** 150 gold
- **Defeat:** 50 gold
- **Draw:** 75 gold

Upgrade Costs (varies by rarity):

Rarity	Level 1→2	Level 2→3
Common	200 gold	500 gold
Rare	400 gold	1000 gold
Epic	800 gold	2000 gold
Legendary	1500 gold	4000 gold

4.1.4. Visual Indicators

- Card rarity shown by border color:
 - Common: Gray/White
 - Rare: Blue
 - Epic: Purple
 - Legendary: Orange/Gold
- Card level shown with stars or numbers (e.g., ★★★ for Level 3)
- Upgrade screen displays:
 - Current stats
 - Next level stats (preview)
 - Upgrade cost
 - Player's current gold balance

4.1.5. Persistence

The following must be saved and loaded:

- Player's total gold amount
- Level of each of the 28 cards
- Gold earned history (for statistics)

5. Feature 4: Challenge Mode

5.1. Functional Requirements

Create 5 pre-designed challenge scenarios with specific objectives and constraints.

5.1.1. The 5 Challenges

Each challenge has unique rules and victory conditions:

1. **“Swarm Master”**

- You can only use swarm troop cards (Skeletons, Goblins, Spear Goblins, Archers, Minions, Minion Horde, Barbarians)
- Must include at least 5 swarm cards in your deck
- Objective: Win normally
- Reward: 250 gold

2. **“Spell Barrage”**

- Your deck must contain all 4 spell cards (Zap, Arrows, Fireball, Rocket)
- Objective: Win the match
- Spells cost 1 less Elixir (minimum 1)
- Reward: 300 gold

3. **“No Buildings Allowed”**

- Cannot use any building cards
- Deck must contain only troops and spells
- Objective: Win normally
- Reward: 200 gold

4. **“Budget Battle”**

- Can only use cards costing 3 Elixir or less
- Deck must have 8 valid cards
- Objective: Win normally
- Reward: 250 gold

5. **“Tank Rush”**

- Can only use high-HP units: Giant, Knight, Valkyrie, Mini P.E.K.K.A, Barbarians
- No spells or buildings
- Objective: Win normally
- Reward: 300 gold

5.1.2. Challenge Progression

- Challenges must be completed in order (unlock Challenge 2 after completing Challenge 1)
- Each challenge can be attempted unlimited times
- Track completion status for each challenge (completed/not completed)
- Implement a star rating system (1-3 stars)
 - 1 Star: Complete the challenge
 - 2 Stars: Complete within time limit (varies per challenge)
 - 3 Stars: Complete with perfect conditions (no damage taken, or very fast completion)
- Track basic performance for each challenge (attempts, completions)

5.1.3. Challenge UI

Challenge Selection Screen:

- List all 5 challenges
- Show lock icon for locked challenges
- Show completion status
- Display challenge name, description, and reward
- “Start Challenge” button for unlocked challenges

During Challenge:

- Display challenge name at top of screen
- Show special rules as reminder (e.g., “Swarm units only!”)
- Track challenge-specific objectives (e.g., time remaining, damage taken)

5.1.4. Design Pattern Considerations

Challenge Mode naturally lends itself to several design patterns:

- **Strategy Pattern:** Each challenge can be a different strategy with unique deck validation and rule enforcement
- **Template Method Pattern:** Define a common challenge flow with hooks for challenge-specific behavior

- **Factory Pattern:** Create challenge instances based on challenge type
- **Decorator Pattern:** Apply challenge-specific modifications (e.g., Elixir cost reduction for Spell Barrage)

Document your chosen pattern(s) in your design pattern report with justification.

6. Feature 5: Quest and Achievement System

6.1. Functional Requirements

Implement a system that tracks player actions and rewards them for completing quests and achievements.

6.1.1. Daily Quests

Overview:

- 3 random quests assigned each day (resets every 24 hours)
- Quests are automatically assigned when the game is launched
- Completing a quest grants gold reward
- Uncompleted quests are discarded at reset

Quest Types (15 possible quests):

1. “Win 3 matches” - 250 gold
2. “Destroy 5 Crown Towers” - 200 gold
3. “Play 10 spell cards” - 150 gold
4. “Deploy 15 troop cards” - 175 gold
5. “Spend 100 total Elixir” - 100 gold
6. “Win a match without losing a Crown Tower” - 300 gold
7. “Play 5 building cards” - 150 gold
8. “Deal 3000 damage with spells” - 200 gold
9. “Win using only common cards” - 250 gold
10. “Complete 2 challenges” - 300 gold
11. “Win a network multiplayer match” - 200 gold

12. “Play 20 cards in a single match” - 150 gold
13. “Win 2 matches in a row” - 300 gold
14. “Destroy an enemy King Tower” - 350 gold
15. “Win a PvP match” - 200 gold

6.1.2. Permanent Achievements

Overview:

- One-time rewards for reaching milestones
- Track lifetime statistics
- Achievements remain locked until conditions are met
- Higher rewards than daily quests

Achievement List (12 achievements):

1. **”First Blood”** - Win your first match (500 gold)
2. **”Tower Hunter”** - Destroy 50 Crown Towers total (750 gold)
3. **”Challenge Master”** - Complete all 5 challenges (1500 gold)
4. **”Three-Star Hero”** - Get 3 stars on any challenge (600 gold)
5. **”Legendary Collector”** - Upgrade a Legendary card to Level 3 (1000 gold)
6. **”Network Warrior”** - Win 10 network multiplayer matches (800 gold)
7. **”Army Builder”** - Deploy 100 swarm troops total (700 gold)
8. **”Spell Master”** - Deal 10,000 damage with spells total (800 gold)
9. **”Gold Hoarder”** - Accumulate 5,000 total gold earned (500 gold)
10. **”Veteran Player”** - Play 50 matches (600 gold)
11. **”Combo Expert”** - Trigger 25 card combos (see Feature 6) (750 gold)
12. **”Undefeated”** - Win 5 matches in a row (1000 gold)

6.1.3. Quest/Achievement UI

Main Screen:

- Menu option: "Quests & Achievements"
- Tab view with two tabs: "Daily Quests" and "Achievements"
- Show notification badge if any quest/achievement is completed but not claimed

Daily Quests Tab:

- List 3 current quests
- Progress bar for each quest (e.g., "5/10 spells played")
- "Claim Reward" button (enabled when completed)
- Countdown timer showing time until reset

Achievements Tab:

- Scrollable list of all achievements
- Locked achievements shown with grayed out icon and "???" description hint
- Unlocked achievements shown with description and progress
- Completed achievements shown with checkmark and "Claimed" status

In-Game Notification:

- When quest/achievement is completed, show popup notification
- Display: "Quest Complete! +250 Gold"
- Can be dismissed or auto-closes after 3 seconds

6.1.4. Persistence

Save the following data:

- Current daily quests and their progress
- Last reset timestamp (for daily reset check)
- All achievement progress and completion status
- Lifetime statistics (total matches, towers destroyed, damage dealt, etc.)

7. Feature 6: Card Combo System

7.1. Functional Requirements

Implement a system that detects specific card combinations and grants bonus effects.

7.1.1. How Combos Work

- A combo is triggered when specific cards are played within 5 seconds of each other
- Combos apply bonus effects to the involved cards/units
- Each combo can only trigger once per 5-second window
- Visual and textual feedback when combo is triggered

7.1.2. The 8 Combo Types

1. "Tank + Support" Combo

- Trigger: Play Giant or Knight, then play any ranged troop (Musketeer, Archers, Spear Goblins, Wizard) within 5 seconds
- Effect: Ranged troop gains +15% damage
- Visual: Gold glow around the ranged unit

2. "Spell Synergy" Combo

- Trigger: Play any spell, then play a different spell within 5 seconds
- Effect: Second spell costs 1 less Elixir (refund 1 Elixir after playing)
- Visual: Sparkle effect when second spell is cast

3. "Swarm Attack" Combo

- Trigger: Play any two swarm cards (Skeletons, Goblins, Spear Goblins, Archers, Minions, Minion Horde, Barbarians) within 5 seconds
- Effect: All swarm units from both cards gain +10% movement speed
- Visual: Speed lines appear on units

4. "Building Defense" Combo

- Trigger: Play any two building cards within 5 seconds
- Effect: Both buildings gain +20% HP
- Visual: Shield icon appears on buildings

5. "Air Assault" Combo

- Trigger: Play Minions and Minion Horde within 5 seconds (either order)
- Effect: All air units gain +15% damage
- Visual: Lightning effect around flying units

6. "Royal Combo" Combo

- Trigger: Play Knight and Archers within 5 seconds (either order)
- Effect: Knight gains +100 HP
- Visual: Crown icon appears above Knight

7. "Siege Mode" Combo

- Trigger: Play Mortar and any defensive building (Cannon, Tesla, Bomb Tower, Inferno Tower) within 5 seconds
- Effect: Mortar range increased by +2 tiles
- Visual: Range indicator circle expands

8. "Rush Attack" Combo

- Trigger: Play Hog Rider and any low-cost card (1-2 Elixir) within 5 seconds
- Effect: Hog Rider speed increased by +20%
- Visual: Dust trail behind Hog Rider

7.1.3. Combo UI Elements

Visual Feedback:

- When combo triggers:
 - Large "COMBO!" text appears in center of screen (1 second duration)
 - Combo name displayed below (e.g., "Tank + Support!")
 - Particle effect on affected units
 - Sound effect (optional but recommended)
- Affected units have visual indicator (glow, icon, effect) for combo duration

Combo Counter:

- Display current combo count during match (e.g., "Combos: 3")
- Shows unique combos triggered (not total triggers)
- Resets each match

Post-Match Rewards:

- Award +10 gold per unique combo triggered in the match
- Display in victory/defeat screen: "Combos Triggered: 4 (+40 gold)"

7.1.4. Combo Detection Logic

Implementation Requirements:

- Track the last 10 cards played with timestamps
- When a card is played, check if it forms a combo with any card played in the last 5 seconds
- Apply combo effects immediately upon detection
- Prevent the same combo from triggering multiple times in the same 5-second window
- Combo effects are permanent for that unit (e.g., +15% damage lasts until unit dies)

Edge Cases:

- If a unit is destroyed before combo effect is applied, effect is wasted
- For spells (Spell Synergy), refund happens immediately after second spell is cast
- Combo effects stack if multiple combos affect the same unit

8. Phase 2 Technical Requirements

8.1. Enhanced Save/Load System

Extend Phase 1's save/load functionality to include:

- Player profile:
 - Player name
 - Total gold balance
 - Lifetime statistics (matches played, wins, losses, towers destroyed, etc.)
- Card progression:
 - Level for each of the 28 cards
 - Total gold spent on upgrades
- Quest and achievement data:
 - Current daily quests and progress
 - Last quest reset timestamp
 - Achievement completion status and progress
- Challenge progress:

- Challenges unlocked
- Stars earned per challenge
- Best times for each challenge

File Format: Continue using the same approach from Phase 1 (Serialization or custom format).

8.2. Network Configuration

Create a configuration file for network settings:

File: `network_config.txt`

```
DEFAULT_PORT=8080
CONNECTION_TIMEOUT=5000
RECONNECT_ATTEMPTS=3
SYNC_INTERVAL=100
MAX_MESSAGE_SIZE=1024
```

Requirements:

- Must be readable and modifiable by users
- Game should load these settings at startup
- Provide default values if file is missing

8.3. Performance Requirements

Same as Phase 1, plus:

- Network multiplayer should have less than 100ms latency on local network
- Game should handle network lag gracefully (show lag indicator, don't freeze)
- Save/load operations for new data should still complete in under 2 seconds
- Combo detection should not cause frame rate drops
- Challenge mode should run as smoothly as normal matches

8.4. Code Quality Requirements

- **Documentation:** All pattern implementations must be documented with:
 - Comments explaining which pattern is used
 - Javadoc for key classes and methods
 - README section describing pattern usage

- **Modularity:** New features should be modular and not require major refactoring of Phase 1 code
- **Testing:** Include unit tests

8.5. Network Configuration File

Include a well-commented `network_config.txt` file with:

- Default settings
- Comments explaining each setting
- Valid value ranges

9. Phase 2 FAQ

Q: Do we have to use specific design patterns?

A: No. The features are designed to naturally lead you toward certain patterns, but you should discover and apply patterns as you solve design problems. Document whatever patterns you use and justify your choices.

Q: Can we modify Phase 1 code?

A: Yes. Good use of patterns should allow you to extend functionality without major rewrites.

Q: How do we test network multiplayer without two computers?

A: Test on localhost first (127.0.0.1). Both host and client can run on the same computer. For final testing, you should test over an actual network.

Q: Can we change the combo effects or challenge designs?

A: Minor adjustments are fine for balance, but major changes *should be discussed with your TA* and then documented and justified in your Challenge Design Document.

Q: Do we need to implement all 15 daily quest types?

A: You should implement at least 10 quest types. Having more variety is better for gameplay.

Q: Can we use external libraries for networking?

A: Basic Java Socket/ServerSocket classes are sufficient and recommended.

Q: How do we handle time zones for daily quest resets?

A: Use the system's local time. Reset quests 24 hours after they were last assigned based on local clock.

10. Final Notes for Phase 2

Key Takeaways:

- Phase 2 is about **software design**, not just adding features

- Design patterns should **emerge naturally** from solving design problems
- **Document your decisions** - explain why you chose each pattern
- **Test incrementally** - don't wait until the end to test features
- **Communicate with your team** - ensure everyone understands the architecture

Remember:

- Start early and work consistently
- Commit to Git frequently with clear messages
- Don't hesitate to refactor if you find better design solutions
- Have fun and learn from the process!

Good luck with Phase 2!